AI Deployment, Vertex AI, Cloud Run & MLOps Complete Study Guide

Introduction to AI Deployment

AI Deployment is the process of taking a trained machine learning model and making it available for real-world use. A model developed in Jupyter Notebook has no business value until it is deployed and accessible to users. Deployment Objectives: • Scalability – Handle thousands or millions of requests. • Reliability – Consistent predictions. • Availability – Accessible 24×7. • Security – Protect data and model endpoints. • Fast Inference – Generate responses quickly. Real-world examples include recommendation systems, chatbots, fraud detection, medical diagnosis assistance, and predictive maintenance.

Why Deployment Matters

Deployment converts research into business value. Benefits: 1. Production-ready AI systems. 2. Real-time predictions. 3. Workflow automation. 4. Better customer experience. 5. Revenue generation. Without deployment, a model remains only an experiment.

Vertex AI

Vertex AI is Google's unified Machine Learning platform on Google Cloud. Main Components: • Data Preparation • Model Training • Hyperparameter Tuning • Model Registry • Endpoint Deployment • Monitoring • Pipeline Automation Advantages: • Managed infrastructure • Auto-scaling • MLOps integration • Security and governance • Support for custom and AutoML models Vertex AI Lifecycle: Data Collection → Training → Validation → Deployment → Monitoring → Retraining.

Cloud Run

Cloud Run is a serverless compute platform for containerized applications. Key Features: • Automatic Scaling • Pay-as-you-use pricing • Container support • HTTPS endpoint generation • Fast deployment Cloud Run is commonly used for: • AI inference APIs • Backend services • Microservices • Event-driven applications Benefits: No server management, low operational overhead, and high scalability.

AI Deployment Workflow

Step 1: Train the model. Step 2: Evaluate model performance. Step 3: Package model inside an API. Step 4: Containerize using Docker. Step 5: Deploy on cloud infrastructure. Step 6: Monitor performance. Step 7: Retrain when performance degrades. Components: AI Model, API Service, Cloud Infrastructure, Monitoring Tools.

Understanding AI Hallucinations

Hallucinations occur when AI generates false or misleading information confidently. Causes: • Insufficient context • Poor quality training data • Ambiguous prompts • Knowledge gaps • Incorrect reasoning Examples: A chatbot inventing references, statistics, or events that never existed.

Reducing Hallucinations

Techniques: 1. Retrieval-Augmented Generation (RAG) Connects LLMs to trusted knowledge sources. 2. Prompt Engineering Clear instructions reduce ambiguity. 3. Human Validation Experts verify important outputs. 4. Reliable Knowledge Sources Use authoritative databases and documentation. 5. Fine-tuning and Feedback Improve model behavior using validated examples.

Monitoring AI Performance

Monitoring ensures that deployed models continue performing correctly. Important Metrics: • Accuracy • Precision • Recall • F1 Score • Latency • Throughput • CPU Usage • Memory Usage Monitoring detects: • Data Drift • Model Drift • Infrastructure Failures • Security Issues

Performance Optimization

Optimization techniques improve speed and reduce costs. 1. Model Quantization Reduce model size using lower precision numbers. 2. Caching Store frequent responses. 3. Resource Optimization Efficient CPU, GPU and RAM usage. 4. Batch Processing Process multiple requests together. 5. Efficient Prompting Use concise prompts to reduce token usage.

MLOps

MLOps combines Machine Learning, DevOps, and Data Engineering. Core Principles: • Automation • Reproducibility • Collaboration • Continuous Improvement Pipeline: Data → Training → Testing → Deployment → Monitoring → Retraining Benefits: • Faster deployment cycles • Reduced operational risk • Better governance • Improved team collaboration

AI Governance

AI Governance ensures responsible and compliant AI use. Key Areas: • Fairness • Transparency • Explainability • Privacy • Security • Regulatory Compliance Organizations use governance frameworks to reduce ethical and legal risks.

Interview & Viva Questions

1. What is AI Deployment? 2. What is Vertex AI? 3. Difference between Cloud Run and Vertex AI? 4. What are AI hallucinations? 5. Explain RAG. 6. What is MLOps? 7. What is model drift? 8. Why is monitoring important? 9. What is quantization? 10. Explain auto-scaling.

Exam Notes

Definitions: AI Deployment – Making trained models available to users. Vertex AI – Google's end-to-end ML platform. Cloud Run – Serverless container deployment platform. Hallucination – AI-generated incorrect information. MLOps – Operational practices for managing ML lifecycle. Very Important Topics: • Deployment Workflow • Vertex AI Features • Cloud Run Architecture • Hallucinations and RAG • Monitoring Metrics • Performance Optimization • AI Governance

Quick Revision Sheet

AI Deployment = Trained Model + API + Cloud + Monitoring.
Vertex AI = Training + Deployment + Monitoring + MLOps.
Cloud Run = Serverless Containers.
Hallucinations = Confident but wrong outputs.
RAG = External knowledge retrieval for accurate responses.
Monitoring Metrics = Accuracy, Latency, Throughput, CPU, Memory.
Optimization = Quantization, Caching, Efficient Prompting.
MLOps = Automating the ML lifecycle.